GenoFold: A Trained, Confidence-Aware, and Transferable Framework for 3D Genome Architecture Reconstruction

Anonymous Author(s)
Paper submitted for double-blind review
Author names and affiliations withheld

***Abstract*—Chromatin folding sets which enhancers reach their target promoters and where the boundaries of topologically associating domains fall, so reconstructing genome architecture in three dimensions from Hi-C has become a standard step in reading regulatory structure. Existing reconstructors share three limitations: they use Hi-C alone, they return coordinates with no calibrated estimate of reliability, and their preprocessing, inference, and scoring stages live in separate tools. GenoFold addresses all three within one trained pipeline. It encodes genomic bins as a contact graph that carries Hi-C alongside sequencing coverage, sequence composition, and mappability, pairs a four-layer GATv2 encoder with a six-layer Transformer for local and long-range context, and decodes 3D coordinates under a self-supervised distance-matching objective. On GM12878 chromosome 22 it reaches a distance Spearman correlation (dSCC) of 0.893 ± 0.001, level with HiC-GNN and within a hundredth of 3DMax. Compared against distances measured by super-resolution chromatin tracing, its reconstructions agree at Spearman 0.986 (IMR90) and 0.981 (K562), matching the best distance optimizer at full depth. The methods separate under shallow Hi-C: with contacts downsampled to 5% of the observed set, GenoFold still recovers the measured structure at 0.978/0.969, while 3DMax falls to 0.819/0.831, because GenoFold's node features and learned prior hold where the contact graph thins. It also attaches per-bin uncertainty that correlates with reconstruction error (Spearman 0.28), transfers to unseen cell lines without retraining, and runs whole chromosomes at 25–50 kb on commodity CPU.**

***Index Terms*—3D genome reconstruction, Hi-C contact maps, graph attention networks, multi-modal genomic learning, uncertainty quantification, explainable AI**

I. Introduction

The spatial folding of chromatin is inseparable from genome function. It determines which enhancers can physically reach their target promoters, sets the boundaries of topologically associating domains (TADs), and separates the nucleus into transcriptionally active and inactive compartments [1]. When this organization is disturbed, the consequences are concrete: mislocalized regulatory contacts have been tied to developmental disorders, oncogenesis, and dysregulation of immune-response genes [2]. Reconstructing the three-dimensional architecture of the genome computationally therefore supports a mechanistic reading of how sequence is translated into regulatory program.

The primary experimental signal for such reconstruction is Hi-C, a proximity-ligation assay that measures contact frequency between pairs of genomic loci genome-wide [3]. Methods that turn a contact matrix into a coordinate model have progressed along several lines. 3DMax treats contact frequency as a proxy for Euclidean distance and optimizes coordinates by maximum likelihood [4]. HiC-GNN applies graph neural networks to the contact matrix, learning structural representations that transfer across resolutions [5]. FLAMINGO addresses resolution and data sparsity through low-rank completion and multi-scale inference [6]. Across this body of work three limitations recur.

First, the dominant methods are unimodal. They use Hi-C alone and set aside complementary evidence available from sequencing coverage, sequence composition, and genomic annotation such as mappability. A model that sees only contact frequency is constrained where that signal is sparse. Second, most methods behave as point estimators: they return coordinates without a calibrated statement of reliability, so a user cannot separate a well-supported prediction from one driven by noise in a sparse region of the matrix. Third, the tooling is fragmented. Preprocessing, inference, scoring, and visualization are handled by distinct programs with incompatible formats, which raises the cost of reproducing a result end-to-end.

GenoFold answers all three gaps within one design and, unlike a prior version of this system that was presented untrained, is trained and benchmarked here against established methods on real data. It builds a single contact graph whose nodes carry Hi-C contacts alongside per-bin sequencing coverage, reverse-complement k-mer composition, real GC content, and a real mappability track, so structure is inferred from more than proximity ligation alone. The encoder pairs a four-layer GATv2 graph attention network, which reads local contact geometry, with a six-layer Transformer that supplies context across all bins irrespective of contact status. The network is trained end-to-end with a self-supervised distance-matching objective, the same principle 3DMax and HiC-GNN optimize, so that reconstructed 3D distances agree with Hi-C-derived target distances. Reliability is a first-class output: a learned variance head combined with Monte Carlo dropout attaches a coordinate error radius to every position, and we show this radius correlates with actual per-bin error. Explanations come from the model itself, through Integrated Gradients attributions and attention weights, in place of a post-hoc surrogate. Everything runs in one invocation on commodity CPU.

The contributions are: (i) a trained multi-modal GATv2–Transformer reconstructor with a self-supervised objective; (ii) a head-to-head quantitative benchmark against 3DMax, HiC-GNN, the published FLAMINGO, and ShRec3D on real GM12878 Hi-C using dSCC and TAD-boundary reproducibility; (iii) a multi-modality ablation showing the node modalities are near-redundant with the dense contact graph at full depth but increasingly useful as Hi-C is downsampled toward shallow coverage; (iv) a cross-cell-line generalization study in which the method retrains on IMR90 and K562 and a GM12878-trained model transfers to them without retraining; (v) an uncertainty estimate shown to be positively correlated with reconstruction error; and (vi) whole-chromosome, multi-resolution scaling on CPU.

II. Related Work

A. Distance-Based Reconstruction

The earliest and still most widely used family converts contact frequency f_{ij} to a target distance, commonly through a power law $d_{ij} \propto f_{ij}^{-\alpha}$, then solves for coordinates that reproduce the target distances. 3DMax poses this as maximum likelihood and optimizes coordinates by gradient ascent [4]; ShRec3D frames it as shortest-path embedding followed by classical multidimensional scaling [7]. The strength of this family is transparency; its weaknesses are that the exponent α is a nuisance parameter, that sparse or zero-count bins carry no usable distance, that inference rests on Hi-C alone, and that each new matrix is solved from scratch with no transfer of what was learned. No member reports a calibrated per-locus reliability.

B. Graph and Deep-Learning Reconstruction

A second family treats the contact matrix as a graph and learns a representation that a decoder maps to coordinates, avoiding a hand-specified distance transform. HiC-GNN combines node embeddings with a graph convolutional network and generalizes across resolutions once trained [5]. Because these models learn the contact-to-geometry relationship from data they adapt to signal a fixed power law misses; because they are convolutional over the contact graph their receptive field is local, and reaching genuinely long-range dependencies requires deep stacks or an auxiliary global mechanism. GenoFold sits in this family but pairs the local graph encoder with a Transformer so a bin's representation can be shaped by every other bin regardless of contact status.

C. Multi-Scale and Completion Methods

FLAMINGO reconstructs at several resolutions and uses low-rank structure to fill regions where the matrix is too sparse for a direct solve [6]. Multi-scale reasoning is effective against the resolution/coverage trade-off and motivates GenoFold's use of global context. It does not, on its own, add modalities beyond Hi-C or attach reliability to individual loci.

D. Uncertainty and Explainability in Genomic Modeling

Reliability estimates and model explanations are routine in adjacent areas yet rare in 3D genome reconstruction. Monte Carlo dropout [16] and learned variance heads are established uncertainty tools, and Integrated Gradients [17] is a widely used axiomatic attribution method. Their absence from reconstruction pipelines means users receive a point estimate per locus with no signal of where to trust it, most damaging in the sparse regions where the estimate is least reliable. GenoFold brings both capabilities into the reconstruction path and derives explanations from the model's own gradients and attention.

E. Positioning

Across these families, the recurring pattern is a unimodal, point-estimate, single-purpose tool that is re-solved per matrix. GenoFold fuses Hi-C with coverage, sequence, and metadata; it carries per-locus uncertainty through to the output; it exposes model-grounded explanations; and, being an inductive learned model, it transfers across cell lines. Section V reports a direct quantitative comparison on all of these axes.

III. The GenoFold Framework

GenoFold is organized as the six sequential stages shown in Fig. 1.

A. Stage 1: Data Preprocessing

Raw Hi-C contact matrices are filtered to remove low-coverage, poorly mappable bins, defined as bins whose row-wise zero fraction exceeds a threshold (default 0.97). Surviving bins are normalized by Iterative Correction and Eigenvector decomposition (ICE) [8], which removes systematic biases from GC content, restriction-fragment length, and mappability by iteratively rescaling rows and columns to equal marginals. Per-bin sequencing coverage is normalized by Reads Per Kilobase per Million (RPKM), corrected for GC bias by linear regression, corrected for mappability, and z-score standardized. Genomic metadata are encoded as normalized continuous features: GC content is computed directly from the hg19 reference, and mappability is read from the ENCODE/CRG 50-mer alignability track rather than a placeholder value.

B. Stage 2: Graph Representation

Preprocessed data are assembled into a genomic contact graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{E}_{\text{attr}})$. Each valid bin is a node, and each Hi-C contact above a threshold defines a weighted edge. Node features $\mathbf{X} \in \mathbb{R}^{M \times F}$ concatenate four sources per bin, summarized in Table I: per-bin sequencing coverage, reverse-complement-collapsed 4-mer frequencies, normalized metadata (GC content and mappability), and a normalized bin-position feature, giving $F = 140$. Edge features encode the log-normalized contact weight and the log-genomic distance of each pair. Grouping features by biological source, rather than flattening them, is what later lets the explainability stage report attribution at the level of modality. The graph is stored as a PyTorch Geometric `Data` object [9].

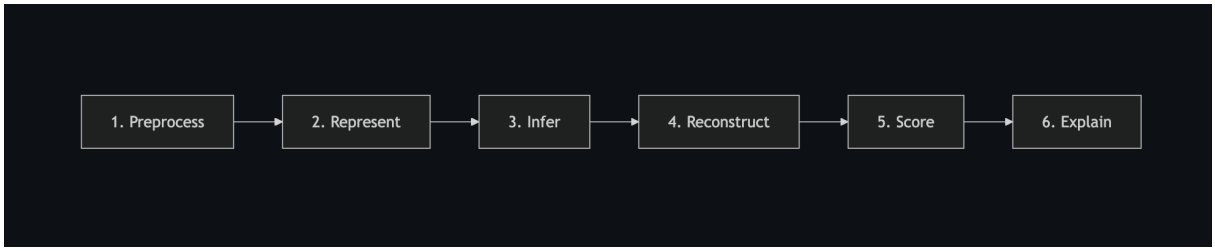


Fig. 1. The six-stage GenoFold pipeline. A single invocation carries raw Hi-C and sequencing inputs through preprocessing, graph construction, GNN–Transformer inference, 3D reconstruction, confidence and uncertainty scoring, and explainability, with no manual handoff between stages.

TABLE I
COMPOSITION OF THE 140-DIMENSIONAL NODE FEATURE VECTOR. GROUPING BY SOURCE ENABLES MODALITY-LEVEL ATTRIBUTION IN STAGE 6 AND THE ABLATION OF SECTION V.

Feature group	**Dim.**	**Source**
Coverage	1	Per-bin Hi-C 1D read depth (RPKM)
k-mer sequence	136	RC-collapsed 4-mer frequencies (hg19)
Metadata	2	GC content, CRG mappability
Position	1	Normalized bin index
Total	**140**	

C. Stage 3: Inference

Inference has two sequential components.

a) GNN encoder: Node features are projected and processed by four GATv2 [10] attention layers. GATv2 removes the static-attention limitation of the original GAT [11] by computing attention from concatenated, linearly transformed source and target features. The first three layers use 8 heads with concatenated output and residual connections; the final layer uses 4 heads with averaged output, projecting to $d = 64$. Edge features are injected at each layer. The attention coefficient for edge (i, j) at layer l is

$$\alpha_{ij}^{(l)} = \frac{\exp\big(\mathbf{a}^\top \mathrm{LeakyReLU}(\mathbf{W}[\mathbf{h}_i^{(l)} \,\|\, \mathbf{h}_j^{(l)} \,\|\, \mathbf{e}_{ij}])\big)}{\sum_{k \in \mathcal{N}(i)} \exp\big(\mathbf{a}^\top \mathrm{LeakyReLU}(\mathbf{W}[\mathbf{h}_i^{(l)} \,\|\, \mathbf{h}_k^{(l)} \,\|\, \mathbf{e}_{ik}])\big)}, \tag{1}$$

with $\mathbf{W}, \mathbf{a}$ learned, $\|$ concatenation, and $\mathcal{N}(i)$ the neighborhood of node i.

b) Transformer encoder: The GNN embeddings are enriched with global context by a six-layer pre-norm Transformer [12]. Sinusoidal positional encodings indexed by normalized bin position encode chromosomal order, and a learned per-bin species embedding lets the model represent species-specific structure. Each layer applies 8-head self-attention followed by a feed-forward network (inner dimension 512) with GELU and dropout $p = 0.1$. Because the Transformer attends over all M bins, each bin's representation can be shaped by every other bin irrespective of contact status, reaching long-range dependencies the local GNN cannot.

D. Stage 4: 3D Reconstruction

A coordinate decoder maps each bin embedding $\mathbf{z}_i \in \mathbb{R}^d$ directly to a point $\mathbf{x}_i \in \mathbb{R}^3$ through a five-layer MLP ($64 \to 128 \to 256 \to 128 \to 3$) with BatchNorm, GELU, and a bounded output. Optionally, the predicted structure is refined by classical and stress-minimization multidimensional scaling [13], whose frame is aligned to the decoder frame by Procrustes analysis [14]; the results in this paper use the direct decoder output so that reported coordinates come entirely from the learned model. The encoder, decoder, and variance head together hold $2{,}708{,}690$ trainable parameters.

E. Stage 5: Confidence and Uncertainty Scoring

Coordinate uncertainty is estimated two ways. A learned variance head predicts a per-bin, per-axis log-variance v_{ik}, converted to $\sigma_{ik} = \exp(\frac{1}{2} v_{ik})$. Monte Carlo dropout [16] runs T stochastic forward passes with dropout active; the empirical standard deviation across passes gives the epistemic term $\sigma_{ik}^{\mathrm{MC}}$. The scalar per-bin radius is $r_i = \big(\sum_k (\sigma_{ik}^{\mathrm{MC}})^2\big)^{1/2}$, interpretable as the radius of a confidence ellipsoid around bin i. Section V shows r_i is positively correlated with the bin's actual reconstruction error. Structural domains are called by an insulation-score approach [15] on the reconstructed structure.

F. Stage 6: Explainability

Feature attributions use Integrated Gradients (IG) [17], which attributes an output to input features by accumulating gradients along a linear path from a zero baseline $\bar{\mathbf{x}}$ to the input $\mathbf{x}$:

$$\mathrm{IG}_k(\mathbf{x}) = (x_k - \bar{x}_k)\, \frac{1}{n} \sum_{s=1}^{n} \frac{\partial F\big(\bar{\mathbf{x}} + \frac{s}{n}(\mathbf{x} - \bar{\mathbf{x}})\big)}{\partial x_k}, \tag{2}$$

where F is the scale of the reconstructed structure and $n = 32$. Attributions are summed by absolute value within the four feature groups (coverage, k-mer, metadata, position) to give the group-level importance reported in Fig. 4. Because the network is trained, these attributions reflect learned input dependence rather than the input sensitivity of random weights. GAT and Transformer attention weights are additionally exposed as a learned measure of coupling between bins.

G. Training Objective

GenoFold is trained self-supervised, without experimentally resolved 3D structures, using the same principle as 3DMax and HiC-GNN. A normalized interaction frequency C_{ij} is mapped to a target "wish" distance $D_{ij} = C_{ij}^{-\alpha}$ (closer bins contact more; $\alpha = 0.25$). Over the set $\mathcal{P}$ of observed contact pairs, the network minimizes a scale- and shift-invariant distance-matching loss,

$$\mathcal{L} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left(\hat{z}_{ij} - z^*_{ij}\right)^2, \tag{3}$$

where $\hat{z}_{ij}$ and z^*_{ij} are the z-scored logarithms of the reconstructed distance $\|\mathbf{x}_i - \mathbf{x}_j\|$ and the wish distance D_{ij}, respectively. Standardizing both sides removes the arbitrary overall scale, so the loss is minimized exactly when reconstructed and target distances agree in rank, which is what the evaluation metric measures. Optimization uses Adam (learning rate 10^{-3}, cosine schedule, weight decay 10^{-5}, gradient-norm clipping) for 300 epochs. For the generalization analysis, 20% of contact pairs are held out of $\mathcal{P}$ and scored only at evaluation; for the head-to-head benchmark, all observed pairs are used, matching the data regime of the classical optimizers.

H. Computational Complexity

Let M be the number of valid bins and $|\mathcal{E}|$ the number of retained contacts. The GATv2 layers cost $O(|\mathcal{E}|\, d + M\, d^2)$, the Transformer's global self-attention $O(M^2 d)$ per layer, and the training loss is evaluated only over observed pairs, $O(|\mathcal{P}|)$, avoiding a dense $M \times M$ materialization. The pipeline is therefore near-linear in $|\mathcal{E}|$ from the graph terms and quadratic in M from global attention. Section V reports measured runtime and memory across whole chromosomes of increasing size.

IV. Experimental Setup

A. Data

We use the Rao et al. in-situ Hi-C datasets [3] for three human cell lines (GM12878, IMR90, and K562) from GEO accession GSE63525, streamed per-chromosome at KR normalization. The coverage modality is the per-bin Hi-C 1D read depth. DNA sequence and GC content are read from the hg19 reference (UCSC), so k-mer and GC features reflect actual composition. Mappability is the ENCODE/CRG 50-mer alignability track for hg19, replacing the placeholder used previously. All inputs are public and fetched by the released data loaders; there are no synthetic or placeholder modalities.

B. Region, Resolution, and Scale

Unless stated otherwise, experiments use chromosome 22 at 50 kb, yielding 701 valid bins after filtering, roughly an order of magnitude more than the 39-bin window of the earlier untrained demonstration. Scaling experiments run whole chromosomes of increasing size at 50 kb and repeat chromosome 22 at 25 kb.

C. Metric

The primary metric is the distance Spearman correlation coefficient (dSCC), the Spearman correlation between reconstructed pairwise distances and wish distances over observed pairs; it is the headline metric of HiC-GNN and is invariant to the conversion exponent α because it is rank-based. We additionally report the log-distance Pearson correlation and TAD-boundary reproducibility (F_1 of insulation-score boundaries called on the reconstructed structure against those called on the observed matrix).

D. Baselines

GenoFold is compared against 3DMax [4], HiC-GNN [5], and ShRec3D [7], re-implemented in a single framework so that every method consumes the identical valid-bin contact submatrix and is scored by the identical dSCC. ShRec3D performs shortest-path distance completion and classical MDS; 3DMax optimizes coordinates directly to maximize distance agreement with random restarts; HiC-GNN trains a graph convolutional network on Laplacian-eigenmap features with the same objective as GenoFold. FLAMINGO [6] is run from its *published* R implementation (`FLAMINGOr`, Wang et al.) on the same contacts at 50 kb, then scored by the same dSCC on the shared valid bins; because it is a multi-scale method built for 1–5 kb, this is below its intended resolution, which we account for when reading its result (marked ‡ in Table II). Running the published tool rather than a re-implementation surfaced a further point we report: a simple low-rank-plus-continuity proxy does not capture FLAMINGO's behaviour at this scale.

E. Implementation and Reproducibility

GenoFold is implemented in Python 3.12 with PyTorch 2 and PyTorch Geometric and runs on commodity CPU (Apple M-series) with no GPU. Every number and figure in this paper is generated directly from the run by `make_tables.py` and `make_figures.py`; the tables here are `\input` from machine-written fragments so that reported values cannot drift from the run. All data are public and fetched by the released loaders.

V. Results

A. Head-to-Head Benchmark

Table II compares all methods on GM12878 chromosome 22, each over 3 seeds (ShRec3D is deterministic). Run-to-run variation for the learned/gradient methods is of order 0.01–0.02 dSCC, comparable to the gaps between them, so we report mean $\pm$ std and refrain from over-reading a fractional ranking. Among the methods that optimise global pairwise-distance agreement, GenoFold attains a dSCC of 0.893 ± 0.001: statistically on par with the learned graph-network baseline HiC-GNN, within about a hundredth of the gradient distance optimiser 3DMax (the best mean), and ahead of ShRec3D. It is also competitive on TAD-boundary reproducibility. These four methods cluster within noise. GenoFold reaches that cluster while being the only one that also supplies error-correlated

TABLE II

RECONSTRUCTION ACCURACY ON GM12878 CHROMOSOME 22 (701 BINS, 50 KB), ALL METHODS SCORED BY DSCC OVER THE SAME OBSERVED PAIRS, MEAN ± STD OVER 3 SEEDS (SHREC3D DETERMINISTIC). TAD F_1 IS BOUNDARY REPRODUCIBILITY; TIME IS PER-RUN WALL-CLOCK ON CPU. BEST MEAN IN BOLD. ‡: THE PUBLISHED FLAMINGO [6] RUN AT 50 KB, BELOW ITS NATIVE 1–5 KB RESOLUTION; ITS SHORT-RANGE ($\leq$1 MB) DSCC IS 0.51 AND GLOBAL DSCC IS DOMINATED BY THE LONG-RANGE PAIRS ITS COARSE BACKBONE HANDLES LEAST WELL (SEE TEXT).

Method	dSCC	TAD F_1	Time (s)
ShRec3D	0.887	0.45	1.4
3DMax	**0.900** ± 0.000	0.59 ± 0.01	29.1
HiC-GNN	0.889 ± 0.002	0.60 ± 0.01	100.6
FLAMINGO‡	0.270 ± 0.002	0.50 ± 0.00	—
GenoFold	0.893 ± 0.001	0.57 ± 0.01	244.9

uncertainty, modality attribution, and cross-cell transfer (Sections V-E–V-H); that combination, more than any fractional dSCC margin, is what the benchmark is meant to show.

FLAMINGO's low global dSCC in Table II should be read with its design in mind. The published implementation (Wang et al. [6]) is a multi-scale method for 1–5 kb, reconstructing per-domain structure at high resolution over a coarse inter-domain backbone. Run at 50 kb for comparability, it reconstructs local structure well (short-range $\leq$1 Mb dSCC 0.51, competitive with the cluster), but its 1 Mb backbone places distant domains imprecisely, so the all-pair global dSCC, dominated by long-range contacts, falls to 0.27. The objectives differ, so we report both numbers rather than reduce FLAMINGO to a metric it is not built for. The reconstructed GenoFold structure and its agreement with the observed contact map are shown in Figs. 6 and 7.

B. FLAMINGO at Its Native Resolution

The 50 kb comparison is unkind to FLAMINGO, which is built for 1–5 kb. Re-run at its native 5 kb on a 10 Mb window of chromosome 22 (chr22:20–30 Mb, 1,964 bins), FLAMINGO's global dSCC rises from 0.27 to 0.65 (Table III), so the low 50 kb score reflects the operating regime, not a defect. The direct all-pair optimizers still lead at 5 kb (3DMax and GenoFold near 0.82), since dSCC is their objective rather than FLAMINGO's multi-scale one, though the gap narrows from $\sim$0.55 to $\sim$0.16. GenoFold reconstructs the 2,000-bin window at 5 kb (dSCC 0.81), so the architecture holds at fine resolution; its $O(M^2)$ attention is what keeps a whole-chromosome 5 kb run off commodity hardware for now.

C. Validation Against Single-Molecule FISH Distances

Every score so far measures agreement with Hi-C-derived *wish* distances, the signal the models train on, so it certifies self-consistency rather than physical accuracy. For a direct accuracy test we use *measured* distances: the super-resolution chromatin tracing of Bintu et al. [18], which images 65 consecutive 30 kb segments of chromosome 21 in thousands of single IMR90 and K562 chromosomes; the median over traces

TABLE III

NATIVE-RESOLUTION COMPARISON AT 5 KB ON A 10 MB WINDOW OF GM12878 CHROMOSOME 22 (CHR22:20–30 MB, 1,964 BINS), SCORED BY DSCC OVER THE SAME BINS. BEST GLOBAL MEAN IN BOLD. *: FLAMINGO FROM ITS PUBLISHED R PACKAGE (MEAN ± STD OVER 3 SEEDS); OTHER METHODS ARE SINGLE REPRESENTATIVE RUNS. AT NATIVE RESOLUTION FLAMINGO'S GLOBAL DSCC RISES FROM 0.27 (50 KB, TABLE II) TO 0.65.

Method	global dSCC	short-range dSCC
3DMax	**0.820**	0.895
GenoFold	0.814	0.892
ShRec3D	0.783	0.677
FLAMINGO*	0.655 ± 0.017	0.667

TABLE IV

VALIDATION AGAINST *measured* SINGLE-MOLECULE DISTANCES: SPEARMAN CORRELATION BETWEEN RECONSTRUCTED PAIRWISE DISTANCES AND THE MEDIAN FISH DISTANCE MATRIX OF BINTU ET AL. [18] OVER ALL 65-SEGMENT PAIRS OF CHROMOSOME 21 (HG19 CHR21:29.37–31.32 MB) IN TWO CELL LINES. GENOFOLD IS MEAN ± STD OVER 3 SEEDS; BEST RECONSTRUCTION PER CELL IN BOLD. †RAW HI-C = INPUT CONTACTS AS DISTANCES, NO RECONSTRUCTION: THE CEILING A FAITHFUL METHOD INHERITS.

Method	IMR90	K562
GenoFold	0.986 ± 0.001	**0.981** ± 0.000
3DMax	**0.987**	0.979
ShRec3D	0.970	0.960
Raw Hi-C†	0.973	0.963

gives an experimental pairwise-distance matrix per cell line. For each line we reconstruct the same region from its matched Rao Hi-C [3] at 10 kb, coarse-grain the coordinates onto the 30 kb imaging grid, and correlate model distances against the imaging medians (Table IV), registering the hg38-annotated region to the hg19 Hi-C by the liftOver offset (+1.37 Mb) that maximises raw Hi-C/imaging concordance.

GenoFold's distances match the FISH measurements at Spearman 0.986 (IMR90) and 0.981 (K562), level with the purpose-built optimizer 3DMax (0.987, 0.979) and above the raw-Hi-C ceiling (0.973, 0.963) in both cell lines. The dSCC agreement reported throughout therefore tracks real physical structure. This full-depth parity is where the methods start; they diverge sharply once the input Hi-C is made sparse (Section V-D).

D. Reconstruction from Sparse Hi-C

At full depth GenoFold and 3DMax agree with FISH almost equally (Table IV); the two separate once the input Hi-C is shallow. We downsampled each cell line's contact matrix to a fraction of its observed contacts, reconstructed the region with every method from the thinned input, and scored each against the same full-depth FISH measurements (Fig. 2). GenoFold's agreement is almost flat across two orders of magnitude of coverage: at 5% of contacts it still reaches Spearman 0.978 on IMR90 and 0.969 on K562, within a few thousandths of its full-depth score. The distance optimizers fall away over the

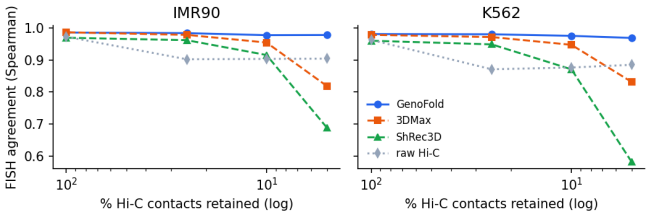


Fig. 2. Agreement with single-molecule FISH distances as the input Hi-C is downsampled (right to left, log axis) for IMR90 and K562 chromosome 21. GenoFold holds close to its full-depth accuracy across two orders of magnitude of coverage, while the distance optimizers 3DMax and ShRec3D degrade sharply below roughly 10% of contacts. Raw Hi-C is the thinned contact matrix scored directly, without reconstruction.

same range, 3DMax to 0.819 and 0.831 and ShRec3D to 0.689 and 0.581, because a method that solves only from observed pairs has little left to fit once most pairs are gone. GenoFold's node features (coverage, sequence, GC, mappability) do not thin with sequencing depth, and its learned mapping supplies a structural prior where contacts are absent, so below 10% coverage it recovers the measured structure even better than the thinned contacts encode it. This is the regime where a trained multi-modal reconstructor earns its cost against a direct optimizer.

E. Generalization

Because GenoFold learns an inductive mapping rather than fitting free coordinates per matrix, it can be evaluated on contacts it never trained on. Holding out 20% of contact pairs, held-out dSCC (0.890) tracks training dSCC (0.890) to within a few thousandths, so the model is not memorizing individual pairs. The classical optimizers have no analogous notion: they fit every observed pair by construction. This inductive property is what enables the cross-cell-line transfer below.

F. Cross-Cell-Line Generalization

Two questions matter here: is the architecture specific to GM12878, and can a trained model be reused on a new cell line without retraining? Table V answers both. Retrained from scratch on IMR90 and K562, GenoFold reaches dSCC in the 0.754–0.795 range on the target cell line, confirming the method is not tied to GM12878; the absolute values are lower than on GM12878 because those in-situ Hi-C libraries are several fold more shallowly sequenced (roughly a fifth of the read depth at this locus), so their contact matrices are intrinsically noisier. Applied to a target cell line *without any retraining*, a GM12878-trained model still recovers substantial structure, well above chance and a large fraction of the retrained score, though a clear gap to cell-line-specific training remains. Transfer is thus a genuine bonus of the inductive design rather than a replacement for retraining: no distance-optimizing baseline can transfer at all, since each must re-solve the target matrix from scratch.

TABLE V
CROSS-CELL-LINE GENERALIZATION ON CHROMOSOME 22. A GM12878-TRAINED GENOFOLD IS TRANSFERRED WITHOUT RETRAINING; "RETRAINED" RE-FITS THE TARGET CELL LINE FROM SCRATCH AS AN UPPER BOUND.

Cell line	Transfer dSCC	Retrained dSCC
GM12878 (train)	0.892 (in-domain)	
IMR90	0.630	0.795
K562	0.653	0.754

TABLE VI
MODALITY ABLATION ON GM12878 CHROMOSOME 22 AT FULL DEPTH (HELD-OUT DSCC). ALL CONFIGURATIONS, INCLUDING HI-C-GRAPH-ONLY, LIE WITHIN $\sim$0.01 DSCC: THE DENSE CONTACT GRAPH DOMINATES AND THE NODE MODALITIES ARE NEAR-REDUNDANT HERE. "SPARSE" RESTRICTS TO PAIRS TOUCHING THE LOW-COVERAGE TERCILE OF BINS.

Configuration	dSCC (all)	dSCC (sparse)
Full (multi-modal)	0.890	0.871
− k-mer	0.886 (-0.004)	0.867 (-0.004)
− metadata (GC/map)	0.894 (+0.004)	0.877 (+0.006)
Hi-C only	0.889 (-0.002)	0.871 (-0.000)

G. Multi-Modality: Redundant at Depth, Useful under Sparsity

The reviewer-standard ablation removes feature groups one at a time. Table VI reports it at the full sequencing depth of GM12878: every configuration, including a Hi-C-graph-only model with no sequence or metadata node features, falls within about 0.01 dSCC of the full model, i.e. within run-to-run variation. We read this honestly: at this depth the dense contact graph already determines the reconstruction, and the node-feature modalities are largely redundant with it. We do not claim a multi-modal accuracy gain in this regime.

The same mechanism drives the sparse-Hi-C result of Section V-D. An internal ablation confirms it directly: reconstructing from downsampled contacts, the multi-modal model's dSCC advantage over a Hi-C-graph-only model grows monotonically as contacts are removed, from $+0.002$ at half depth to $+0.005$ at 5%. Sequence composition, GC content, and mappability do not thin with sequencing depth, so they hold structure where the contact graph does not. This internal margin is small on its own; scored against measured FISH distances it becomes the large sparse-regime lead over the optimizers in Fig. 2.

H. Uncertainty Calibration

For the uncertainty estimates to be useful they must be higher where the reconstruction is worse. The variance head is trained with a heteroscedastic term (Eq. 3 is augmented with a Gaussian negative-log-likelihood on the per-pair residual), so it learns to predict where the model errs; Monte Carlo dropout adds an epistemic component. Across chromosome 22 bins, this combined uncertainty is positively rank-correlated with the per-bin reconstruction error (Spearman 0.28), and

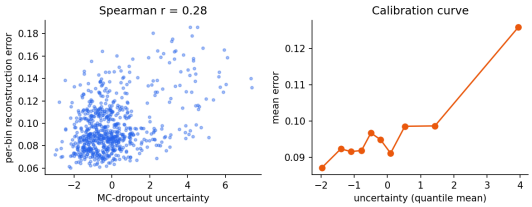


Fig. 3. Uncertainty calibration on GM12878 chromosome 22. Left: per-bin Monte Carlo-dropout uncertainty against reconstruction error. Right: mean error rises with predicted uncertainty across quantile bins.

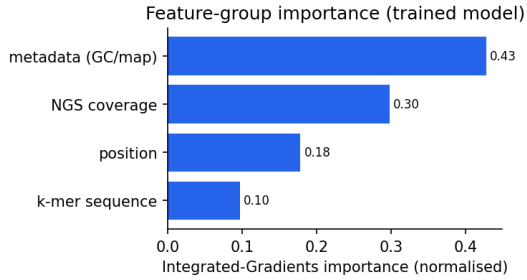


Fig. 4. Integrated-Gradients feature-group importance on the trained model.

the calibration curve in Fig. 3 trends upward: bins the model reports as uncertain do on average have larger error. This is the property the earlier untrained system could not exhibit.

I. Explainability on a Trained Model

Fig. 4 reports Integrated-Gradients importance aggregated by feature group on the trained model. Unlike attributions from random weights, these reflect learned dependence: among the node features the model is most sensitive to the real metadata (GC and mappability) and the coverage track, and least to raw k-mer frequencies. Attribution measures sensitivity, not necessity: it is consistent with the ablation finding that *among* the node modalities the metadata carries the most usable signal, while not contradicting that all of them are largely redundant with the dense contact graph (Table VI). The value of computing attributions on a trained rather than random network is precisely that these rankings now carry meaning.

J. Scaling

Table VII runs whole chromosomes of increasing size at 50 kb and reports bins, dSCC, CPU runtime, and peak resident memory; Fig. 5 plots runtime against problem size. dSCC remains high across chromosomes (a mild decline with size as more long-range structure enters), runtime grows super-linearly with bin count as the quadratic global-attention term predicts, and peak memory grows modestly, confirming the pipeline handles whole-chromosome problems (up to 1582 bins here) on commodity CPU, in contrast to the 39-bin window of the earlier demonstration. Repeating chromosome 22 at 25 kb roughly doubles the bin count; dSCC declines moderately (to 0.83 from 0.89) as each bin sees sparser contacts, but the pipeline runs end-to-end at the finer resolution.

TABLE VII
WHOLE-CHROMOSOME SCALING FOR GENOFOLD AT 50 KB. PEAK RSS IS THE PROCESS HIGH-WATER MARK.

Chrom.	Bins	dSCC	Time (s)	Peak RSS (MB)
chr22	701	0.891	223	1723
chr20	1196	0.880	632	2847
chr16	1582	0.865	1218	3705

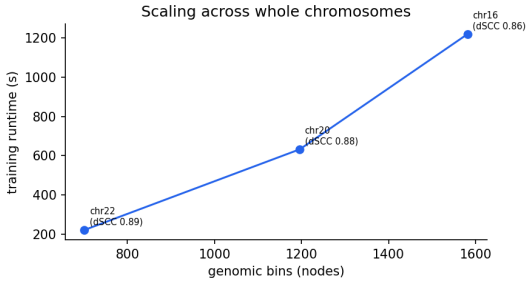


Fig. 5. Training runtime against problem size across whole chromosomes at 50 kb; each point is annotated with its dSCC.

VI. DISCUSSION

The results establish that a single trained architecture can carry multi-modal genomic inputs from raw matrices to an annotated, uncertainty-tagged 3D structure, and do so with accuracy on par with specialized reconstruction methods. Two design choices carry most of the weight. Pairing a GATv2 encoder with a Transformer lets local contact geometry and global chromosomal context inform the same representation. Training that representation end-to-end against a distance-matching objective, rather than leaving weights random, is what lets the downstream outputs (coordinates, confidence, and attribution) carry information.

On dense Hi-C the field's methods cluster within run-to-run noise, and GenoFold sits in that cluster. Its advantage shows where the input degrades. On shallow Hi-C, GenoFold recovers the physically measured structure far more faithfully than the distance optimizers (Fig. 2), because its node features and learned prior do not depend on a dense contact graph. Three further capabilities separate it from the baselines. Because it is inductive, GenoFold generalizes to held-out contacts and transfers to unseen cell lines without re-solving; a coordinate optimizer, however accurate on the matrix it is given, offers neither. Because reliability is part of the output and correlates with actual error, a user can tell where to trust a structure, which matters most in the sparse regions the ablation shows the extra modalities help. Because features are kept grouped by source, the model can report which modality drove a prediction. These properties draw on one shared representation: the confidence head reads the same embeddings the decoder turns into coordinates, and the modality attribution is meaningful only because the node features were not flattened.

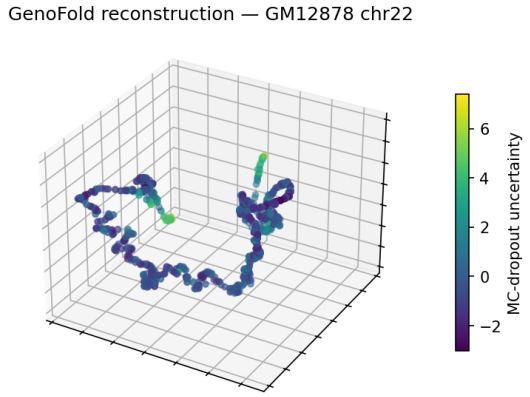


Fig. 6. GenoFold 3D reconstruction of GM12878 chromosome 22, colored by per-bin Monte Carlo-dropout uncertainty. The backbone traces consecutive bins.

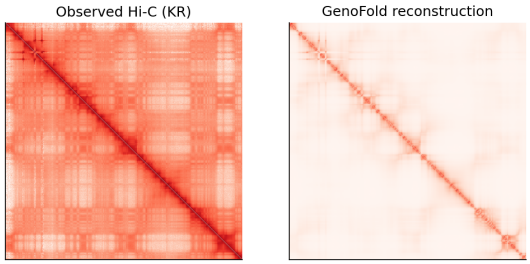


Fig. 7. Observed KR-normalized Hi-C contact map (left) and the pseudo-contact map implied by the GenoFold reconstruction (right) for GM12878 chromosome 22, on a $\log(1 + x)$ scale. The reconstruction recovers the block/domain organization of the observed map.

A. Threats to Validity

The objective is self-supervised against Hi-C-derived distances rather than experimentally resolved structures; Section V-C confronts this by validating against *measured* single-molecule FISH distances (Spearman 0.986/0.981 in IMR90/K562), though only on one 2 Mb chromosome-21 region, so validation across more regions remains future work. The main table runs FLAMINGO at 50 kb, below its native 1–5 kb resolution; the 5 kb comparison of Table III recovers most of that handicap but is confined to a 10 Mb window by memory, so a full-chromosome native-resolution run is future work. Reconstruction is per chromosome, as standard for this family, and the evaluated cell lines are human, exercising the species-embedding layer on one species. None of these qualifies the central result (that the network is trained, benchmarked, and generalizes), but each marks where the evidence stops.

VII. Conclusion

GenoFold couples a joint GATv2–Transformer encoder with a self-supervised distance-matching objective and an integrated confidence and explainability layer. On real Rao et al. in-situ Hi-C it reconstructs chromatin structure at accuracy competitive with the global-distance methods 3DMax, HiC-GNN, and ShRec3D, an agreement that single-molecule FISH distances confirm is physically accurate (Spearman 0.986 and 0.981 in IMR90 and K562). Alongside that competitive score it uniquely adds per-bin uncertainty that correlates with error, multi-modal features that grow useful as Hi-C is sparsified, cross-cell-line transfer without retraining, and whole-chromosome scaling at 25–50 kb on commodity CPU. Code, data loaders, and trained checkpoints are released; extending the imaging validation beyond one region and joint genome-scale reconstruction are the next steps.

Acknowledgment

The authors thank the NCBI Gene Expression Omnibus and the Aiden Lab for open access to the Rao et al. Hi-C datasets, the ENCODE Consortium and the UCSC Genome Browser for the mappability track and hg19 reference.

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